

9.2

a

$$E(X) = 1/\lambda = 2$$

$$P(X > 6) \leq \frac{2}{6} = \frac{1}{3}$$

b

$$Var(X) = 1/\lambda^2 = 4$$

$$P(X > 6) = P(X - \mu > 6 - \mu)$$

$$P(X - \mu > 4) \leq P(|X - \mu| > 4)$$

$$P(|X - \mu| > 4) \leq \frac{Var(X)}{4^2} = \frac{1}{4}$$

c

$$P(X > 6) = 1 - P(X \leq 6)$$

$$P(X \leq 6) = \int_0^6 \lambda e^{-\lambda x} dx = 1 - e^{-6\lambda}$$

$$P(X > 6) = e^{-3}$$

9.7

$$E(X_i) = 1000, \sigma = \sqrt{Var(X_i)} = 900$$

$$S = \sigma^2 500_{i=1} X_i$$

$$E(S) = 2500(1000), Var(S) = 2500(900^2)$$

$$P(S > 2500 \text{premium}) = P(S - 2500(1000) > 2500(\text{premium} - 1000))$$

$$P(S - 2500(1000) > 2500(\text{premium} - 1000)) \leq \frac{Var(X)}{(2500(\text{premium} - 1000))^2}$$

$$= \frac{900^2}{2500(\text{premium} - 1000)^2}$$

We want this to be less than 0.001

$$0.001 \geq \frac{900^2}{2500(\text{premium} - 1000)^2}$$

$$\rightarrow (\text{premium} - 1000)^2 \geq 324000$$

$$\rightarrow \text{premium} \geq \sqrt{324000} + 1000$$

For CLT:

$$P(S \leq 2500\text{premium}) = P\left(\frac{S-2500(1000)}{50(900)} \leq \frac{2500(\text{premium}-1000)}{50(900)}\right)$$

$$= \Phi\left(\frac{2500(\text{premium}-1000)}{50(900)}\right) \geq 0.999$$

$$\rightarrow \frac{2500(\text{premium}-1000)}{50(900)} \geq 3.1$$

$$\rightarrow 2500(\text{premium} - 1000) \geq 3.1(50(900))$$

$$\rightarrow \text{premium} \geq 3.1(900)/50 + 1000$$

$$\rightarrow \text{premium} \geq 1055.8$$

9.12

a

$$E(X) = 2, E(Y) = \frac{1}{2}, E(X + Y) = 2 + \frac{1}{2} = \frac{5}{2}$$

$$P(X + Y > 10) = \frac{5/2}{10} = \frac{1}{4}$$

b

$$\text{Var}(X) = 2, \text{Var}(Y) = \frac{1}{12(b-a)^2} = \frac{1}{12}, \text{Var}(X + Y) = 2 + \frac{1}{12} = \frac{25}{12}$$

$$P(X + Y > 10) = P(X + Y - \frac{5}{2} > 10 - \frac{5}{2}) \leq P(|X + Y - \frac{5}{2}| > \frac{15}{2})$$

$$P(|X + Y - \frac{5}{2}| > \frac{15}{2}) \leq \frac{\text{Var}(X+Y)}{\frac{15}{2}^2} = \frac{1}{27}$$

10.4

a

$$p_{X|N}(k|n) = \binom{n}{k} \frac{1}{2}^n, 0 \leq k \leq n \leq 100$$

b

$$\text{For binomial } E(X) = np \text{ so } E(X|N = n) = \frac{n}{2}$$

c

$E(X) = \sum_{n=0}^1 00 \frac{n}{2} (p_N(n)) = \frac{E(N)}{2}$, from binomial distribution $E(N) = np = 25$

Therefore $E(X) = \frac{25}{2}$

10.6

a

$$f_Y = \int_0^y \frac{x+y}{4} dx = \frac{y^2}{8} + \frac{y^2}{4} = \frac{3y^2}{8}$$

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{2}{3} \frac{x+y}{y^2}, 0 < x < y < 2$$

b

$$P(X < \frac{1}{2} | Y = 1) = \int_0^{\frac{1}{2}} f_{X|Y}(x, 1) dx = \frac{2}{3} \int_0^{\frac{1}{2}} x + 1 dx$$

$$= \frac{2}{3} (\frac{1}{8} + \frac{1}{2}) = \frac{5}{12}$$

$$P(X < \frac{3}{2} | Y = 1) = \int_0^1 f_{X|Y}(x, 1) dx = \frac{2}{3} \int_0^1 x + 1 dx$$

$$= \frac{2}{3} (\frac{1}{2} + 1) = 1$$

c

$$E(X^2) = \int x^2 f_X(x) dx$$

$$f_X(x) = \int_x^2 \frac{x+y}{4} dy$$

$$f_X(x) = \frac{1}{2} + \frac{1}{2}x - \frac{3}{8}x^2$$

$$E(X^2) = \int_0^2 \frac{1}{2}x^2 + \frac{1}{2}x^3 - \frac{3}{8}x^4 dx$$

$$E(X^2) = \frac{14}{15}$$

On the other side

$$E(X^2 | Y = y) = \frac{2}{3} \int_0^y x^2 \frac{x+y}{y^2} dx = \frac{7}{18} y^2$$

$$\int_0^2 \frac{7}{18} y^2 (\frac{3}{8} y^2) dy = \frac{7}{48} (\frac{2^5}{5}) = \frac{14}{15}$$

They are equivalent!

10.15

a

let C_A denote the number of claims made by policy A holders.

$$P(C_A \geq 5) = 1 - \sum_{i=0}^4 P(C_A = i)$$

Let C denote the total number of claims

$$P(C_A = i) = \sum_{c=0}^{\infty} P(C_A = i | C = c) P(C = c)$$

$$= \sum_{c=i}^{\infty} \binom{c}{i} \left(\frac{1}{4}\right)^i \left(\frac{3}{4}\right)^{c-i} e^{-12} \frac{12^c}{c!}$$

$$= \left(\frac{1}{4}\right)^i e^{-12} \sum_{c=i}^{\infty} \binom{c}{i} \left(\frac{3}{4}\right)^{c-i} \frac{12^c}{c!}$$

$$= \frac{1}{i!} \left(\frac{1}{4}\right)^i e^{-12} \sum_{c=i}^{\infty} \frac{c!}{(c-i)!} \left(\frac{3}{4}\right)^{c-i} \frac{12^c}{c!}$$

$$= \frac{12^i}{i!} \left(\frac{1}{4}\right)^i e^{-12} \sum_{c=i}^{\infty} \frac{1}{(c-i)!} \left(\frac{3}{4}\right)^{c-i} 12^{c-i}$$

$$= \frac{3^i}{i!} e^{-12} \sum_{c=i}^{\infty} \frac{9^{c-i}}{(c-i)!}$$

$$= \frac{3^i}{i!} e^{-12} e^9$$

$$= \frac{3^i}{i!} e^{-3}, C_A \sim \text{Poisson}(3)$$

$$P(C_A \geq 5) = 1 - \sum_{i=0}^4 \frac{3^i}{i!} e^{-3}$$

b

Let C_B denote the number of claims made by policy B holders

We apply similar work to find that $C_B \sim \text{Poisson}(9)$ (flip $c - i$ and i in the binomial expression)

$$\text{Thus } P(C_B \geq 5) = 1 - \sum_{i=0}^4 \frac{9^i}{i!} e^{-9}$$

c

$$C \sim \text{Poisson}(12)$$

$$P(C \geq 10) = 1 - \sum_{k=0}^9 \frac{12^k}{k!} e^{-12}$$

10.38

a

$$f_Y(y) = e^{-y}, f_{X|Y}(x|y) = ye^{-yx}$$

$$f_{X,Y}(x,y) = ye^{-y(x+1)}$$

b

$$f_{Y|X} = \frac{f_{X,Y}}{f_X}$$

$$f_X = \int f_{X,Y} dy = \int_0^\infty ye^{-(x+1)y} dy = \frac{1}{(x+1)^2} \text{ integral calculator}$$

$$f_{Y|X} = (x+1)^2 ye^{-(x+1)y}$$

Knowing $X = x, Y \sim \Gamma(\lambda = (x+1), \alpha = 2)$